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Implementing HACCP systems in Europe

Kerry Ingredients

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7.1 Introduction

Originally founded in 1974 as a dairy farmers cooperative in the south-west of Ireland, the Kerry Group is now a leader in global food ingredients markets, and a leading producer of consumer foods for selected European markets. Established as a public company in 1986 from head offices in Tralee, County Kerry, the Group has achieved sustained profitable growth and pursued a strategic global expansion programme. This has led to the establishment of manufacturing, technical and marketing facilities in Ireland, the UK, France, Italy, Germany, the Netherlands, Poland, Hungary, the USA, Canada, Mexico, Brazil, Australia, New Zealand and Malaysia. The Group currently has over 12,000 employees world-wide, and sales in 1998 were IR£1.73 billion with pre-tax profits of IR£93.9 million.[1]

The Kerry Group is split into three operating divisions: Ingredients, Foods and Agribusiness.

The Ingredients division contributes 63.5% of sales turnover with world-wide activities across a broad product range including savoury ingredients, coating systems, snack seasonings, bakery ingredients, fruit preparations and dairy ingredients. Products are sold either directly to industrial food manufacturers or through wholesale outlets and agents to caterers and smaller users. As a global supplier, Kerry Ingredients is ideally placed to support major multinational customers, while maintaining the flexibility to supply small-scale and more specialist customers locally.

The Foods division contributes 33.6% of sales turnover and is more focused on the European market. The product range includes recipe dishes, cured meats, savoury pastry products, sausages, prepared poultry, flour, baking mixes,

spreads, dairy products, juices and mineral water. Products are either sold through major retail outlets, or distributed by a van sales fleet to caterers and smaller retailers. Kerry Foods occupies the number one branded position in the majority of its existing markets and is a major supplier of customers' own-brand products.

The Agribusiness division contributes 2.9% of sales turnover and maintains the link with agricultural producers in Ireland. Feedstuffs are supplied and milk is purchased and collected for use in the production of dairy ingredients. This special relationship with primary suppliers is echoed throughout the business.

Because the HACCP systems that I have been working on are relatively immature, the information in this chapter will reflect experiences gained during the initial stages, up to and including methods of implementation. It will include important lessons learned and things I would have done differently with the benefit of hindsight.

7.2 HACCP and business strategy

It hardly needs to be stated that food safety is the top priority for the Kerry Group, as it is for all responsible food manufacturers. All Kerry manufacturing units have HACCP systems in place which are regularly reviewed to ensure that they continue to be effective.

One common misconception I have found, however, is that many people believe that HACCP is a stand-alone system that will in itself ensure the production of safe food. While it is true that HACCP plays a very important role, it is only one of the elements of an effective food safety management system. Figure 7.1 illustrates this in the form of a 'food safety house' where the foundations represent prerequisite systems, the walls represent HACCP systems and controls and the roof represents general quality systems including traceability and recall procedures.

The HACCP plan is similar to a house layout plan: it identifies which of the 'walls' are 'load-bearing', where they must be built and how strong they need to be. But the house plan itself does not build the walls. Nor does it ensure that they have been built correctly. It is just the starting point. This is also true of the HACCP plan, which details the controls that are required to produce safe food, but these only become effective when HACCP has been properly implemented.

Everybody knows that the walls of a house must be built on solid foundations. HACCP too must be built on solid foundations in the form of effective prerequisite systems such as supplier quality assurance (SQA), good manufacturing practices (GMPs), personnel hygiene and cleaning.

Furthermore, the walls of a house are of little use without a waterproof roof over them. HACCP must be covered by an effective quality system, which is founded on the main structural elements of the 'walls' and protects against exposure to food safety incidents.



Fig. 7.1 The food safety house.

HACCP is of course one of the most important elements of food safety management. Just as a structure is not a house if it has no walls, so a management system cannot control food safety if HACCP is not in place.

This analogy has been useful when explaining HACCP to colleagues. A house is something with which they are all familiar. This makes it easier for them to understand food safety management as a total concept. The relative importance of each of the key elements becomes clear. They quickly realise that HACCP must be totally integrated into the management system if food safety controls are to be effective.

7.3 How big is the job?

This is one of the most common questions asked at the outset of a study, along with: What should be done first? What resource will be needed? How long will it take? How much will it cost? How do we go about it?

When building a new factory the answers to these questions are easy: GMP, hygienic design and pest-proofing measures can be built in at the start. The HACCP study can be performed on the proposed process flow diagram, and all necessary food safety control measures can be designed into the process prior to installation. The quality systems can be developed around the CCPs identified in the HACCP plan. SQA and training can take place during the construction phase. In this way food safety will be totally controlled from start-up.

However, in reality most studies are performed on existing factories. In these circumstances, it becomes much more difficult to answer these questions. The reason for this is that every business is different. The size and age of the factory, the product range, the production processes and the people – these combine to make each operation unique. All of the elements of food safety management are likely to be present in some form; it is unlikely that the business would exist if they were not. But the amount of work required to improve the controls will depend on the maturity of the existing systems.

It is again helpful to consider the food safety house analogy, because a HACCP study on an existing system is very much like renovating an existing house. The amount of work required, the time it will take and the estimated cost are totally dependent on the structural condition at the outset. It is impossible to judge the amount of work that will be required on the house unless a detailed survey is performed by a building expert, and a comprehensive list of faults is prepared.

Ideally, the starting point for a HACCP study should be a detailed audit by an expert in food safety. All elements of food safety management should be assessed for their completeness and effectiveness, and a comprehensive list of defects prepared. It will then be possible to judge the scale of the task ahead.

The next stage in the house renovation process is to analyse the survey. Each of the faults must be considered in turn, and an estimate made of the work required to return the property to sound condition. This will be expressed in terms of materials required and hours of labour involved, so that the estimated cost can be calculated.

Of course, the next stage in the HACCP process is the same. Each of the defects found with the food safety system must be taken through a similar thought process, then materials, labour and cost estimated. It may be necessary to involve an external consultant with relevant experience if this expertise is not available inside the organisation. At this point, it is useful to map out a rough action list, which itemises each of the defects. Blank columns can be left for the work required headed ‘materials’, ‘hours of labour’ and ‘estimated total cost’. Columns should also be left for ‘priority’ and ‘responsibility’. This can then be used as a check-list to ensure that all necessary information is gathered to prepare a project plan.

7.4 Prioritisation in the project plan

It is normally necessary to introduce an element of prioritisation. As with house renovation, there will rarely be a job list that is short enough for all work to be completed immediately and which is easily affordable. There will be constraints such as budget and availability of skilled labour, which must be considered.

There will also be a natural order to the work. It is normal with house renovation to complete structural repair work on the foundations first, then the walls, and finally the roof. There is no point fitting the best roof in the world if

the foundations are about to give way! Of course there will always be exceptions to this: in certain circumstances it may be necessary to perform emergency repair work on the roof to make it completely waterproof before starting work on the walls. It may even be necessary to shore up the walls before it is safe to perform the original structural survey!

This is also true for HACCP, where work on the prerequisite systems will normally take priority. Some emergency repair work on the quality systems may be required early in the project to make sure they are watertight. Some emergency food safety controls may need to be introduced even before the HACCP study is performed (e.g. if there is no metal detector in a process this must be installed immediately!).

Once the action list has been completed, it will be possible to produce a detailed project plan. This must take into account the priority rating, the availability of experienced personnel and the time they can allocate to the project, plus the budgetary considerations previously mentioned. The overall length of the project will be heavily dependent on the number and experience of the people involved. It is possible to plan many of the improvements in prerequisite systems to take place concurrently if there are sufficient people to take individual responsibility for each action. The original list can itself be used as the basis for an action plan to drive the non-HACCP improvements. The HACCP study will normally be planned to start before these reparations have been fully completed.

It is obvious that the actions considered as the highest priorities should be tackled first. However, it is important to realise that the *time scales* allocated to these actions must be consistent with their priority rating. If insufficient resources are available to allow the action points to be completed in a realistic time, then the resource must be increased, not the time scale.

It is usual in the project plan to break the HACCP study down into its key phases. This will not be covered in detail here, as it is well documented elsewhere.[2] The phases will of course be studied in detail later in this chapter.

The role of the project manager is also well documented elsewhere. However, because it is rare to find project management skills and HACCP expertise in the same individual, it is often recommended that the quality representative be trained in project management. An alternative approach is to allocate a skilled project manager to the HACCP team, and to train this person in HACCP techniques. In practice, the engineering representative will probably possess the required project management skills. It is probably a better long-term investment for the business to train the engineers to be food safety experts than it is to train the quality staff to be project managers!

During the study itself and during implementation it is usual to encounter complications and additional problems that could not have been foreseen at the outset. These may require additional resource or expenditure to resolve them within the time scale allowed. A builder will always overestimate the time and cost of a house renovation project because the work rarely goes according to plan. When preparing the HACCP project it is also wise to overestimate the time and cost so that there are no surprises at the end.

The final stage in any house renovation project is the interior decoration. In effect, this is a cosmetic exercise which has no effect on the structure, but adds a comfort factor that few of us would be without. When implementing a food safety system into a factory, the primary objective is to make sure that the controls actually work in practice. Making it look pretty is important, but it is best to leave the tidying up until the end.

Just as routine maintenance on the newly renovated house is necessary to stop it deteriorating to its former level, routine maintenance is also necessary on the food safety system to ensure that it stays effective.

More importantly, if further major modifications are to be made to the house, their effect on the structural integrity must be assessed *before* the work takes place. The careless removal of a load-bearing wall could cause the whole house to collapse. This is also true with HACCP. If any major changes in manufacturing methods are proposed, it is essential to reassess the safety implications *before* the change takes place, or it may result in a catastrophic failure of the food safety system!

Finally, external forces such as earthquakes or hurricanes may affect the structural integrity of a house. In the same way exterior influences such as BSE or food scares may challenge the integrity of a food safety system: if the system is well enough designed, it should withstand all foreseeable external forces.

7.5 The Polish experience

7.5.1 Kerry Pegromar, Poland

One of the subdivisions of Kerry Ingredients produces processed fruit products for the dairy, bakery and catering markets, with manufacturing facilities throughout Europe. Much of the soft fruit is sourced from our own factory in Poland. Kerry Pegromar is one of the most modern fruit freezing operations in eastern Europe. Based in Kielce, two and a half hours south of Warsaw, it is in the heart of the soft-fruit growing area of the country.

Individually quick frozen (IQF) strawberries are the mainstay of the business, with several thousand tonnes being frozen in the 20 or so days of the season. Other major IQF products include sour cherries, rhubarb, blackcurrants, redcurrants, gooseberries and blueberries. Total quantity of fruit frozen is over 10,000 tonnes per year. Frozen fruit is also supplied directly to non-Kerry customers throughout Europe.

The processing operation is relatively simple: fruit is graded on receipt, stored for a limited period under refrigeration, washed, pre-sorted, quick frozen in continuous blast freezers, further sorted to customer specification, packed, metal detected and placed in frozen storage prior to dispatch.

Because large volumes of raw material are processed in a very short time, most of the fruit is placed into intermediate frozen storage as ‘semi product’ which undergoes final sorting to customer specification as required through the year.

My involvement with Pegromar began in 1997: my brief was to spend 50% of my time on site helping to implement food safety controls that would be acceptable to UK customers.

At this time food safety systems based on HACCP had already been introduced for key fruit types, although these were relatively immature. SQA systems were well established, with training programmes in place to communicate the required quality standards for the fruit to the suppliers. Traceability was possible back to supplier and day of delivery. Key quality systems were in place and functional, although they were not approved to ISO 9000 standard.

7.5.2 Pegromar project plan

In order to assess the status of the systems on site we decided to employ an external consultant to perform a baseline audit of the factory. This generated a list of actions required to improve food safety controls.

The main areas of concern were GMP, cleaning, pest control, glass auditing and HACCP coverage. A detailed action plan was prepared using a very simple Lotus Approach database: this software allowed the list to be presented in a number of different formats depending on the target audience. A general action list with a brief description of the work required was useful to communicate the scale of the overall task to senior management. More detailed individual action plans were prepared for each responsible person: initially these were limited to the high priority actions so that the length of the list would not overwhelm the recipient. Lower priority actions were then gradually added to the individual action plans as initial tasks were completed.

This method worked well, and was flexible enough to cope with late additions to the list as unforeseen problems were discovered. Improvements highlighted during customer audits could also be included with ease.

7.5.3 HACCP in a foreign language

The HACCP team consisted of the QA manager, the production manager, the engineering manager and me. The team was relatively inexperienced in HACCP techniques, as at the time there was little reference information in Polish and no recognised training courses that they could attend. The original Campden HACCP document had been translated into Polish, and was being used as a guide.[3] I decided it would not be possible for me to train the team formally, and that it would be necessary to lead them through the actual study and to let them learn by experience.

Language was a major barrier, as I did not speak Polish, and the rest of the team did not speak English except for the QA manager, who spoke only a little. It was obviously necessary to include an interpreter in the team so that we could communicate with each other. This caused a number of problems during the hazard analysis stage. First, the meetings took at least twice as long as they would in a single language because everything had to be said twice: once in

English, then once in Polish, and vice versa for the response. Second, because the interpreter was non-technical, it was necessary to look up many of the words in a dictionary, which took up still more time. Third, of course some of the meaning was lost in the translation, which led to much confusion which needed to be resolved through further explanation and discussion. The process flow diagrams proved to be extremely useful communication tools because they were relatively easy to translate into Polish, and provided a common reference point that all team members could understand (see Fig. 7.2).

7.5.4 The HACCP study

The HACCP study itself was undertaken in a number of logical stages. Because of the seasonality of the fruit harvest, when I arrived in Poland in September 1997 there was very little fresh fruit being processed. The new season would not start until May 1998. There was therefore little point starting the study with the arrival of the fresh fruit, as it was already 'semi product' in the freezer. So we decided to divide the study into two main stages, post-freezing and pre-freezing, with the aim of completing both stages by the start of the 1998 season.

The initial HACCP work that had been completed by the local management team had covered each fruit type separately. This made sense because every fruit has slightly different inherent hazards. For example, strawberries are relatively low risk in terms of foreign bodies, whereas cherries have an in-built stone that could choke a child, blackcurrants and redcurrants often arrive heavily contaminated with sticks, and gooseberry bushes carry very dangerous thorns. The various fruit types also require slightly different treatment, so flexibility is built into the process to cope with this (e.g. slicing of rhubarb, pitting of cherries, etc.). Specific hazards associated with each process stage must be identified if they are to be controlled. The same method was therefore adopted for the follow-up study, with each fruit being reviewed separately.

I had to get to know the people and the process quickly. It is not possible to perform a HACCP study effectively if you do not understand the production process. One of the best ways to get to know an unfamiliar factory is to construct a process flow diagram. This may take several weeks, depending on the complexity of the process, but it is essential that the flow diagram is accurate in every detail, because this is also the first stage of the HACCP study. If a step is omitted from the diagram, then it will not be assessed at the hazard analysis stage, and therefore the associated hazards will never be controlled.

Hazard analysis is another key stage in the HACCP process. This must be performed diligently so that *all* hazards are identified for all steps in the process. If a hazard is missed at this stage, it cannot be identified as a critical control point. This may mean that the finished product is still unsafe even after HACCP has been implemented. This would obviously be unacceptable.

We started the hazard analysis with the highest risk fruits in terms of sales volume and food safety (i.e. frozen strawberries, cherries and gooseberries). Since most hazards were foreign bodies, the inexperience of the team was not a

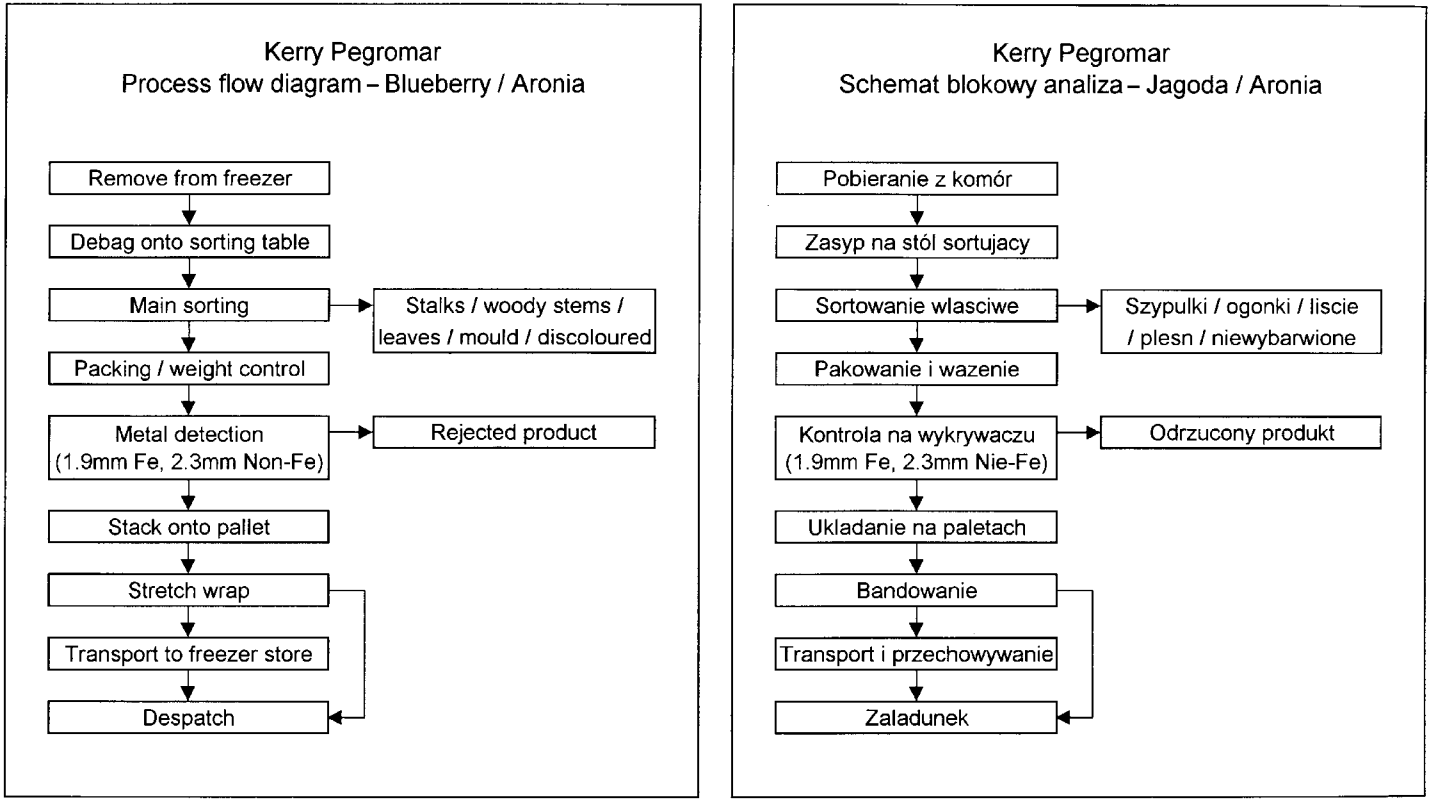


Fig. 7.2 Block flow diagrams in English and Polish were used to aid communication.

major issue; they were easily able to recognise potential food safety issues of this nature. However, on the microbiological front, their inexperience led to a tendency for both spoilage organisms and pathogens to be raised as hazards. This was easily resolved, because points where spoilage alone was concerned were either identified at the time or were removed later during the CCP identification.

Once the team had completed the hazard analysis, it was possible for me to overcome some of the language difficulties by performing the majority of the CCP identification on my own with the help of the CCP decision tree. A decision matrix was used to record the path through the tree for each hazard in case any of the decisions were questioned later. The team was then brought together to review and approve the CCPs. Once this was completed for the three primary fruits, a HACCP control chart was constructed. Although the original intention was to prepare separate control charts for each fruit, in reality the CCPs after the freezing stage were so similar that it made more sense to draft a single control chart for all fruits post-freezing. This was confirmed as the correct approach when the hazard analysis and CCP identification had been completed for the secondary fruits post-freezing, as they too could be covered on the same chart.

The original control chart was prepared in English, and had to be translated into Polish for the team to review and agree the controls for each CCP. Once this was done it was for the rest of the team to complete the implementation and training, as it could only be done in Polish. This was enormously frustrating, because implementation is the final key stage, and I could not influence its progress directly. Worse still, I could not communicate with the trainees to check their level of understanding. All I could do was to audit the results, and to steer the team through to successful implementation.

The same procedure was followed for the raw fruit prior to freezing, except that we could not verify this part of the process flow diagram until the fruit started to arrive. It was therefore necessary to perform the HACCP review on the provisional process flow diagram, because the controls had to be in place *before* the season started. Then as soon as each fruit type began to arrive we verified the flow diagram to make sure it was accurate. When discrepancies were found we revised the flow diagram accordingly, and reviewed the hazards to make sure we had not missed anything. Thankfully, this was a rare occurrence.

The CCPs were included in the quality system so that they would be effectively implemented. This was relatively straightforward, as none of the controls was particularly complicated, and the majority of them were already in place. The quality system for frozen fruit and vegetables was subsequently approved to ISO 9002 by Lloyds Register Quality Assurance in January 1999.

Although we achieved the objectives that we were set, this could not be described as a ‘model’ project. One of the fundamental requirements of HACCP is good communication. It would have been far simpler if I had spoken Polish, if the team had spoken English, or if they could have been trained in their natural language, then given more freedom to progress on their own. It would not be wise to follow this example unless, as we found, there is no realistic alternative.

7.6 Expansion of the Polish operation

7.6.1 Project outline

In January 1998, the decision was made to install a process line at Pegromar to produce pasteurised fruit preparations for the rapidly expanding yoghurt and ice cream markets in eastern Europe. I was the technical representative on the project team from the beginning, and was able to ensure that food safety considerations were built into the design from the start.

It was decided to build the new process area after the frozen storage stage. This would maintain both logical flow of fruit through the factory and maximum distance from the raw fruit to the pasteurisation process to minimise the risk of yeast and mould contamination.

Logical flow *within* the process area was also a major consideration, along with hygienic design and pest proofing of the processing halls.

For the process, the best design features were taken from sister factories in the UK and France, and combined to make a system that would be at the leading edge of fruit processing technology. But a conscious decision was also made to keep the process as simple as possible. Food safety controls that would virtually eliminate the risk of product contamination by cleaning chemicals were designed into the process, along with features that would ensure that critical processing temperatures were accurate and were automatically recorded.

7.6.2 The HACCP study

A HACCP study was performed on the proposed process flow diagram prior to installation. On this occasion, it made no sense to perform the study from first principles, as none of the team members, myself included, had sufficient expertise in fruit processing for the team approach to be effective. Instead, the existing HACCP plans were taken from sister factories in the UK and France. These were thoroughly evaluated to confirm the similarities with the proposed process and product range. More importantly, the differences had to be highlighted and all associated hazards and CCPs identified and built into the modified HACCP plan. This method worked perfectly well: it stopped us repeating mistakes made elsewhere in the group and allowed us to benefit from their experience. It also saved a lot of time and effort on our part.

Performing HACCP prior to the installation of a new process is probably the most powerful use of this technique. It identifies all critical stages in the process, including those that have been overlooked during conception, and enables them to be designed into the process and installed prior to start-up. It also identifies the process parameters that are critical to food safety so that the most appropriate monitoring equipment can be specified and used from the start. Finally, it highlights the manual process records that are required, so that these can be built into the quality system from day one. It is therefore possible to start production with total confidence in the ability of the process to make safe food.

7.6.3 Operator training

No matter how well a process is designed, it is almost impossible to make it 'operator proof'. Like it or not, the production operators play a significant part in the achievement of finished product quality and food safety standards. It is therefore vital that they know what they are doing and understand the implications of their actions.

Training of operators is often one of the weakest links in the food safety chain. Rarely is sufficient time or resource allocated to the task, and seldom is it adequately completed. To help understand why, think for a moment about learning to drive a car. First, trainees have to learn what all the controls do, then they have to learn the sequence required to make the car move. Next, they practise these control sequences under supervision while driving on the road. At this stage it is necessary for them to think of each action in the sequence before it is completed, making it difficult for them to think about driving safely, so they drive very slowly at first. When they get past this stage they no longer have to think about which gear they need to be in, or which pedal to press. They can then start to concentrate on driving safely, so that they do not cause an accident, which may put someone's life at risk. At this stage they must pass a test to prove that they have mastered the basic requirements needed to go solo.

When we train operators to use a new machine or new process, it is just like teaching them to drive a car. They are not fully trained until running the process equipment is second nature to them. When they have reached this level of awareness, they can concentrate on thinking about controlling food safety, so that lives are not put at risk.

This may be considered by some to be a poor analogy, because it suggests that we want to produce unthinking trainees. Quite the opposite is true – we want to make sure that the trainees are not allowed to 'go solo' or to 'drive' too fast until they are able to control the process so that food safety is never compromised.

The selection and training of operators was to be a crucial factor at Pegromar, because pasteurisation was new to the factory and the operators had no previous experience of this form of processing. The only sensible course of action was to allow the trainees to experience it at first hand.

Three months prior to the target commissioning date, when the Pegromar process was literally still on the drawing board, we selected the three leading operators from existing staff. They were sent to their sister factory in the UK, for three weeks of hands-on fruit processing. This was not ideal because language was a major barrier, which meant that all training and communication had to be done through a non-specialist interpreter. For this reason, the majority of the basic training took the form of physical demonstrations. During the first day the trainees were introduced to a simple recipe. All of the ingredients were present in their correct proportions, and the role of each one was explained. The principles of pasteurisation and subsequent aseptic controls were described in detail. Then the main food safety and quality

problems were discussed, with examples being shown wherever possible. A number of laboratory-scale samples were prepared in the presence of the trainees to demonstrate the importance of following the recipe. These included several deliberate failures which served to reinforce this message and to further develop their understanding of the product.

This led to bench-scale and then pilot-scale production trials by the operators themselves. When we were confident that they had mastered the basics, the trainees were placed with leading staff in the factory, and spent an average of two working days in each department. This brief immersion gave the Polish staff an insight into the daily routine of a fruit production unit. Obviously, they were not yet fully trained, but at least they had gained some relevant experience.

Several weeks later the management team went through a shorter but very similar training exercise.

Just prior to start-up the operators were returned to the UK for a further two weeks' refresher training, accompanied by the quality control manager. As well as receiving further training himself, the quality control manager also acted as interpreter for the group, and developed a close working relationship with the rest of the team. This established his credibility as a leader, and earned him the respect of the operators.

7.6.4 Commissioning and start-up

The process was commissioned in October 1998, and as soon as it was operational we began to train the staff in earnest. We devised an economical training recipe, which included all of the functional ingredients in quantities that had to be weighed out (i.e. not full bag quantities). One batch of this was made every day, and filled off aseptically into cleaned, sterilised 800 litre transport tanks. Samples taken during preparation and at the point of filling were tested as if they were finished product to make sure that they met the specification. In this way, we completed the training of the original group of operators and their support staff in all aspects of the Pegromar process. Then using this method and their expertise, we were able to train fully the second shift of operators on site in Poland.

The factory received its first customer orders in December 1998, and made its first real product early the following month. At the time of writing, the plant has been in production for four months, and several hundred tonnes of product has been made. There has not been a single microbiological failure or foreign body issue with fruit preparations made at Pegromar.

7.7 Kerry Ingredients, Portbury, UK

7.7.1 Background

In November 1998, I started work at the Portbury factory near Bristol. This is the newest facility in the seasonings and coatings subdivision of Kerry Ingredients.

Commissioned in 1997, this purpose-built factory was constructed on a greenfield site alongside the M5, one of England's main arterial motorways. The objective was to create a state-of-the-art manufacturing unit that would replace the outdated production facility in central Bristol.

The factory makes a range of dry blended seasonings and savoury flavours. Customers include most savoury food manufacturers in the UK, as well as many others in Europe and the Middle East.

The process is relatively simple: ingredients are weighed out, mixed together, sieved, packed, metal detected, stored, then dispatched to customers. It is made complicated by the sheer numbers involved: around 2,700 finished products are made from approximately 1,500 raw materials.

As would be expected with a new factory, GMP and hygienic design were built in from the start. SQA and raw material controls were good, which is essential with a dry blending operation, because the process does not include a pasteurisation step. The quality and safety of the finished product are therefore almost totally dependent on the raw materials used. Quality systems are effective and are approved to ISO 9002.

7.7.2 Review of existing HACCP plan

My brief was to lead the team through a review of the existing HACCP system to make sure it was still current. The system had been put in place before the factory was built, and was therefore based on the planned process flow diagram. Because the original HACCP plan was very complicated, it was decided to start the review from first principles. Foreign bodies are the main food safety concern with this range of products, so we focused on these first. As a rule, it is best to start the HACCP process with foreign bodies if the team members are inexperienced, because these hazards are easiest to relate to.

The team approach was used to produce the foreign body plan. Then because of the relative simplicity of the process and the time constraints on the other team members it was decided that the technical manager and I should perform the microbiological and chemical hazard analysis. The team then met to review this analysis and to perform the CCP identification. The resulting CCPs were added to the foreign body plan to complete the study in early January 1999.

A detailed process flow diagram had been prepared and used for the review: this included all steps in the process, their interrelationships and all inputs and outputs. A much simpler *schematic* process flow diagram was prepared for inclusion in the HACCP plan. This could be easily understood by operators and could be sent to customers without fear of betraying our technology to our competitors.

When the new plan was compared with the original HACCP, it was reassuring to find that nothing new had been identified. The only difference was that several of the foreign body control points from early in the process had been upgraded to full CCP status. This was because we found that the sieve mesh sizes being used for some products could not be relied on to remove larger non-

metallic foreign matter from the product. While still identified as CCPs, the sieves are also used as part of the verification procedure to ensure that the earlier controls are fully effective.

7.7.3 Implementation and training

Implementation of the plan was relatively straightforward because most of the CCPs were already controlled under the original HACCP plan, and were fully integrated into the quality system. A number of additional work instructions were required to reflect the changes in emphasis, which gave goods-in and warehouse staff more responsibility for food safety.

The most important part of the implementation process was the training of the staff. The objective of this 'operator awareness' training was to get the trainees to understand what the controls were and why they were necessary, so that they would not be tempted to take short cuts that could compromise food safety.

During the training, the first thing the trainees were made to realise was that *they* were the potential consumers. This immediately caught their attention and made food safety a personal issue. From then on, it was relatively easy to raise their awareness of the issues and controls by asking and answering a number of simple questions.

When asked which was more important, quality, food safety or production efficiency, the trainees usually came up with 'food safety' as the right answer. This was made clear to them by emphasising that we could kill somebody if we got it wrong, which could lead to closure of the factory, in which case quality and efficiency would be irrelevant.

The question 'What are the three main types of food safety contaminant?' started to focus the group on issues specific to our process. The food safety contaminants were covered on a need-to-know basis. To have given too much information at this stage would probably have been counterproductive as the operators may not have understood it all, so they would have 'switched off'. At Portbury we only blend together dry ingredients. During the discussion on microbiological hazards neither the characteristics of individual pathogens nor their heat destruction was covered. This information would have been irrelevant, because there is minimal risk of microbiological growth, and no 'kill' stage in the process. The information must be kept relevant to the audience, and if possible specific examples from their workplace should be used to illustrate each point made. During the session, we discussed general microbiological hazards relating to raw materials, cross-contamination from dirty equipment or people, and the risks of water in our process which must be avoided to prevent microbiological growth. In other words, we covered the CCPs from the HACCP plan.

Glass and sharp metal were used as examples of foreign bodies to awaken the imagination of the trainees. The limitations of sieves and metal detectors were then covered in detail. The increasing ineffectiveness of sieves larger than 3–4 mm and the inability of metal detectors to pick up small pieces of metal,

including wire and stainless steel swarf, were emphasised. The need for prevention was obvious.

Actual customer complaint statistics relating to foreign bodies were then used to illustrate this. A kind of mini hazard analysis exercise was carried out, where each of the categories was taken in turn, the likely source 'brainstormed' and the preventative measures highlighted. To further emphasise the need for controls, the point was made that the consumer does not normally know that the food is unsafe until after it enters their mouth.

At this stage, the trainees were more than ready to receive some good news. The question 'How can we guarantee food safety?' introduced the subject of HACCP. Again, the information relating to the origins of HACCP was kept as brief and as interesting as possible for the audience. The mental picture of the symptoms of food poisoning in zero gravity was used to good effect to explain why this system was originally developed. The HACCP plan was only mentioned in passing so that the operators knew it had been completed.

They did not need to know every CCP in the process, only those relevant to them, and they had to understand why they were important (if they had not already worked this out for themselves). They needed to know that all CCPs had been identified and were under control, so they could be confident that the product was safe.

They were introduced to the concept of a food safety 'jigsaw puzzle', where all of the pieces must be in place to give a complete 'picture' of food safety. They were told that, as individuals, they were all pieces in that jigsaw puzzle. The schematic process flow diagram was used to show them the overall picture, so that they could see that all of the CCPs were in place. They were then given a pocket-sized laminated card with details of the hazards and controls relevant to their job (see Fig. 7.3). Each of the CCPs was discussed in turn, and the reasons for these controls were reiterated. The operators from each department were trained as a group so that their attention could be focused on controls specific to their area. In most cases, the action to be taken by the operators was designed to either prevent or control the immediate problem. Then if necessary they were to inform their supervisor, so that a management decision could be taken to bring the situation back under control.

The training concluded by considering total quality management and internal supplier/customer relationships, with the emphasis on the importance of supplying safe products and services to internal customers. It was important to leave them with the message that their individual actions could make a real difference to the overall safety of the finished product.

7.7.4 Nonconformity management

One of the most important elements of any food safety system is nonconformity management. This is necessary in case there is a failure at one of the CCPs. However, this part of the system is often poorly developed or badly

Critical Control Points – Packing

CCP	Hazard	Control
10.1	Filler parts falling into product (seals, nuts & bolts, swarf if moving parts work loose).	✓ Inspect filling machine every time it is cleaned. ✓ Make sure all moving parts are present, undamaged & properly fitted.
11	Foreign matter inside packaging before filling	✓ Look out for foreign objects in packaging during filling.
12.1	Failure to detect & reject metal contaminated product.	✓ Check metal detector with test pieces at the start of every batch & end of day.
13	Contamination of our, or our customer's factory with foreign matter from pallet (e.g. glass).	✓ Inspect all pallets before transfer to packing area / use
13.2	Wood / nails from pallet puncture packs.	✓ Use cover sheet on every wooden pallet stacked.

✓ Report all problems to your supervisor

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Fig. 7.3 Plastic laminated 'crib' cards were given to operators.

misunderstood. 'Control of nonconforming product' and 'corrective action' are fundamental requirements of a quality system, and are of course two of the clauses of ISO 9000. It is ironic therefore that ISO 9000 is the main cause of the confusion, because many businesses create two separate systems in order to meet their interpretation of the requirements of the standard. In fact, these two clauses are so intimately linked that they can only be effective if they are managed within a single system.

Portbury were running an 'internal nonconformity sheet', a 'corrective action' sheet and a 'finished product reject' sheet. These three records were replaced by a single 'nonconformity report', which was used in all cases where finished product was affected. This was designed to lead and record the thought processes in the

event of a product or manufacturing failure. It was divided into five sections:

1. To control the problem.
2. To investigate the cause.
3. To deal with the product affected.
4. To take corrective action to prevent its recurrence.
5. To review and close the incident.

This was originally introduced as a paper system, which we soon found to be impossible to manage. The nonconformity sheets would get buried under mounds of paperwork on people's desks, and it was difficult to keep track of the stages of completion. We incorporated it into a networked Lotus Notes 'quality' database, which quickly resolved the management issues and brought significant additional benefits. These included the generation of weekly and monthly summary reports, trend analysis, search options and significantly improved communication to all interested parties.

7.7.5 Verification procedures

Auditing is an important part of the maintenance of an effective food safety system. HACCP audits are often performed independently of other audits. Although there is nothing wrong with this method, it suggests to visiting external auditors that the CCPs are operating independently from the rest of the quality system. If the controls have been fully integrated, there should be no need to audit them separately: the CCPs should automatically be covered during routine quality system or hygiene/housekeeping audits. In practice it is relatively straightforward to achieve this, providing that the quality system is well established and the CCPs are clearly identified within it so that they are not overlooked during the audit.

We plan to achieve this at Portbury by including the HACCP plan in the quality database. It will then be possible to list the CCPs automatically on the appropriate audit checklist, along with outstanding actions from the previous audit, saving time and making the auditor much more effective.

7.8 The benefit of hindsight

Most of the points that follow are related to language and communication issues because the majority of this work was done abroad. Some of them have already been raised in earlier sections but they will be explained in more detail here.

The most important thing would be to make sure at the outset that there is a common understanding of food safety among the senior management team. The 'food safety house' analogy was developed quite late on. If this had been available at the start, it would have been used for this purpose. This would have helped significantly at all stages of the project.

Access to computers was also a major problem. The original intention was to use local computers to create the HACCP documentation, because the local management team needed to have the master version of these documents. This was a big mistake. Apart from the obvious problems of availability of hardware and different makes or versions of software from site to site, there was a major problem with language. When working in a foreign language version, the Windows environment is like an alien world, with familiar icons becoming unrecognisable when 'foreign' text is applied to them. Simple tasks like printing were a nightmare at first, as a dictionary had to be used to look up the words before the commands could be made with confidence, making progress very slow. With hindsight, it is obvious that either the English version of the software should have been installed on the local computer, or better still, a laptop should have been available from the start. I now carry a laptop computer, which has proved invaluable, and has paid for itself many times over. If starting out again I would always insist that a dedicated computer be made available; the choice of desktop or laptop should be made to suit the circumstances.

When working with people in a foreign language it is more likely that progress will be made if both parties know a little of the other's language. Obviously the more fluently the language is spoken, the easier it will be to communicate. Probably the best option for implementing any sort of change in another country is to use local experts who can work in their mother tongue. They will then be able to cover the whole project from conception through to implementation and training. If this is not possible and there is no common language between you and the local management team I would strongly suggest recruiting knowledgeable staff locally who can speak both languages fluently. In my experience, it is very difficult to communicate through an interpreter because too much of the meaning is lost in the translation.

To function effectively, the HACCP team members must either be trained or experienced in the principles and techniques of HACCP. When working abroad, the language issues make it much more difficult to train the staff personally. It would have been extremely useful to be able to send staff on a training course in their own language in the knowledge that they would receive the message that was intended. There has long been a need for a standard training package that is internationally recognised, written in simple language so that it can be translated easily. Perhaps the newly available World Health Organisation training package will meet this need in future.[4]

During HACCP implementation, be it in the UK or abroad, operators also need to be trained so that they achieve a level of understanding of the controls required, but more importantly the reasons for these controls. This can only be done effectively in their natural language. Ideally, a member of the local management team should carry out this training, as it is necessary to draw on examples from their working environment. If done well, it will strengthen the working relationship between the manager and the staff. At Pegromar, when a completely new process was installed in their factory it was necessary to send operators abroad to gain relevant experience. If we had to do this again we

would always send a member of the management team with them. Their role would be to act as interpreter, to make sure that they were properly treated, to ensure that they really understood the process and to develop the working relationships between the new team members.

While working elsewhere in Europe during the same period as the Polish project, similar problems were experienced. However, a number of additional issues were encountered.

In one case, there seemed either to be differences in the legal requirement regarding HACCP or differences in the interpretation of this requirement. The emphasis at this factory was on the preparation of impressive HACCP documentation that would satisfy the authorities, with seemingly little regard given to the implementation of the controls. Of course, this situation was immediately rectified, but readers should be aware that local management might in some cases expect to work to the 'letter' rather than the 'spirit' of the food safety legislation.

Another issue was one of prejudice, where recent problems of BSE in the UK provoked the 'mad cow' response from some people in Europe. In their eyes, the lack of credibility of UK-based food safety specialists meant that they were highly sceptical of all recommendations made. It was therefore difficult to implement the necessary additional controls. Although I do not know the solution, I feel that this problem needs to be raised so that others may be forewarned.

Most worryingly of all, in Europe there seemed to be a general lack of awareness of the allergenic properties of nuts. Obviously, if nuts are not recognised as a food safety issue, then they cannot be identified during hazard analysis, so will not be controlled as CCPs. It is essential that all such situations are identified and controlled so that lives are not put at risk.

With reference to the HACCP plan, a process flow diagram will always form part of this document. In the early days, we made the mistake of including a fully detailed flow diagram in the plan. Now a schematic version of the diagram is always included, which is easier to understand and is much less commercially sensitive if it is sent to customers.

To end on a positive note, we have recently been able to combine many of the innovations that have been used separately elsewhere. The most significant single development has been the networked 'quality database', which has taken off on the back of the information technology revolution. Although this was primarily introduced as a system to manage internal raw material and finished product nonconformities, it is rapidly expanding and will soon include all aspects of quality systems management in Kerry Ingredients. Specifically, the master copies of all documents and work instructions (including HACCP) will be maintained in the database. Internal audits will also be scheduled and reported here, and all associated corrective actions managed as a live prioritised action plan for each factory. Customer complaint investigations will soon also be incorporated. This system is still being developed at the time of writing, but it is planned to roll this out to all UK factories in the Ingredients division in the

near future. The true power of the system will then be realised. It will be possible for anyone, anywhere to view all incidents of raw material nonconformity for all deliveries across the UK. Innovations such as this will eventually make all factories much more efficient. Data that has been held in traditional paper systems will be presented in the form of management information, so that we will finally be able to make informed decisions that will improve the future performance of the business. This will enable us to make the essential transformation from reactive to proactive food safety management.

7.9 References

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